

KUSHAL KRISHNAPPA

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EDUCATION

Northeastern University, Boston

Master of Science in Computer Science (GPA: 4.0/4.0)

Courses: Programming Design Paradigms, Algorithms, DBMS, Web Development

Jan 2025 - Dec 2026

Boston, MA

PES University, Bengaluru

B.Tech in Computer Science and Engineering

Courses: Machine Learning, Network & Information Security, Big Data, Operating Systems

Aug 2018 - Jun 2022

Bengaluru, India

PROFESSIONAL EXPERIENCE

RoonCyber

Software Engineer, Boston, MA

Jan 2026 - Present

- Secured **10+ endpoints** by building a **multi-tenant API auth system** to enable M2M integrations. Implemented key management, RLS-based isolation, and middleware (caching, RBAC, rate-limiting) to accelerate enterprise deal closures.
- Enhanced **eBPF-based security pipeline** with custom header-scanning rules to pinpoint exact vulnerability location in HTTP requests, reducing MTTD by 90% for application-layer security incidents.

Pure Storage (Everpure)

Member of Technical Staff 2, Bengaluru, India

Jan 2023 - Dec 2024

- Owned full SDLC for **Auto Simulator Recovery**, architecting microservices-based distributed systems using Flask, RabbitMQ, Redis, PostgreSQL, Temporal and Nomad. Reduced simulator recovery time from 5 hours to 10 minutes, increasing productivity for over **2,000 test environments** and handling 200+ daily operations with $\geq 95\%$ reliability.
- Designed and built the **Deployment Orchestrator Service** to manage one-click updates across **9,000+ on-prem VMs**, leveraging real-time VM state and enforcing concurrency limits. Prevented cascading failures during rollouts, enhanced VM update reliability and maintained global infrastructure availability $\geq 90\%$.
- Automated VM lifecycle management on Kubernetes using KubeVirt (provisioning, snapshots, teardown) across 1,500+ test environments and migrated the fleet from **Ubuntu 14 to 22 with 100% compatibility** via Packer/Jenkins/**Dominator** workflow.

Mercedes-Benz R&D

Software Engineer, Bengaluru, India

Mar 2022 - Dec 2022

- Built scalable **ETL pipelines** for OTA updates data using PySpark on Databricks with Delta Lake architecture, processing 90TB+ historical and ongoing OTA updates data, enabling comprehensive data visibility for downstream analytics.
- Built OTA analytics dashboard using **Java Spring Boot** and **Vue.js** to transform complex update logs into actionable reports, surfacing update failure patterns that informed rollout decisions and replaced OTA team's manual triage process.
- Strengthened system security by implementing **OAuth 2.0** authentication for OTA services. Automated deployment and scaling of service through Kubernetes in **Azure DevOps CI/CD** pipelines
- Identified **2 high-latency endpoints** and drove resolution during on-call, reducing p95 response time by 50%.

PROJECTS

- Beacon** | Designed and built a **notification service in Go using Temporal**, enabling reliable event-driven email alerts across 2 personal web services; set up SMTP server with Cloudflare DNS and exposed a REST API for service integration.
- Auth Server** | Built a multi-tenant auth service in **Go** using **gRPC** and **Protocol Buffers**, implementing JWT-based auth, token rotation, tenant isolation and low-latency inter-service communication with focus on scalability and security.
- Distributed Key-Value Store** | Implemented in C++ featuring binary write-ahead logging (WAL), with multi-node replication support for high availability and data consistency across cluster nodes.
- dsapanicle.com** | Built and shipped algorithmic challenges platform using **AI-driven development (Claude Code, agentic workflows)**, serving 10-20 active users with full CI/CD via GitHub Actions + Hostinger webhooks.
- Calendar** | Implemented custom **Interval Tree** data structure for **optimized event conflict detection in $O(k + \log n)$** ; built full-stack calendar application using MVC architecture, Java + Swing, design patterns and SOLID principles.

PUBLICATIONS

An Enhanced Deployment of 5G Network Using Multi-Objective Genetic Algorithm - IEEE, 2022 | Implemented a genetic algorithm to optimize dense 5G deployment, minimizing cost while maximizing coverage and efficiency.

TECHNICAL SKILLS

Skills: Go, Python, Java, C++, TypeScript, LLM, Prompt/Context Engineering, Claude Code, Temporal, Docker, MongoDB, PostgreSQL, Redis, Linux, Nomad, Kubernetes, KubeVirt, Ansible, Packer, Terraform, Jenkins, **Dominator**, Prometheus, Grafana, OpenStack, Flask, Django, Spring Boot, Next.js, JUnit, PyTest, Azure, Git, GitHub Actions, Azure EventHubs, CI/CD, Async I/O, Concurrency, Networks, Virtualization, Code Review, RCA, Monitoring, Troubleshooting, OpenTelemetry

Certifications: AWS Fundamentals: Specialization, Hashicorp Terraform Associate (004), Web Application Security Testing with OWASP ZAP, Object Oriented Programming - Advanced (Java OOP), Foundations of Responsible AI Learning